Subject: bio203 (Current Papers)

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Bio203 current papers for final term spring 2018

1) oining DNA fragments with _____by DNA ligase is a relatively efficient process? cohesive ends
2) Chemically synthesized adaptor molecules which have a preformed cohesive end can be used to insert _____foreign DNA into vector? blunt ended
3) RFLP____Random fragment length polymorphism
4) RFLP 2times____ Random fragment length polymorphism
5) STR____short tendun repeats
6) _____ The use of multiple, unique primer sets within a single PCR reaction? MULTIPLEX PCR
7) Unique human _____60% , 100%, 50%, none of these
8) Army of which country used _____Canada, india, US, all of these
9) Assembly PCR is synthesis of long DNA structures by performing PCR on a pool of long oligonucleotides .
10) In sucide is typically used in polegenetics.
11) PCR used to produce a unique fingerprints of amplified product lengths is called (INTER SEQUENCE SPECIFIC PCR)
12) RT pcr is widely used in ----(expression profiling).
13) in nested pcr -----used to amplify a fragment.(.two pairs of primer).
14) mutiplex pcr one reaction by using more than ------(0ne pair of primer).
15) in molecular technique the RPD stand for(.Random Amplified Polymeric DNa).
16) Denaturation temperature:Reduces double stranded molecules to single stranded, 90–96oC, 20-45 seconds.
17) primer may occur due to the Hybridization of primer each other.
18) Taq polymerase most commonly used polymerase and code for (Thermus aqauticus)..
19) two primer sets are used in _____pcr Nested
20) multiplex PCR enables the simultaneous amplification in many targets on interest in …… One reaction Two reaction Three reaction
21) primer dimer occure due to the Hybridization
22) Set of each primers work on the A. high temp b.low temp C. low ph D. room tem
23) At least how many R-M systems are known? 4
24) Molecular paleontology refers to the recovery and analysis of _____.DNA
25) PCR has also facilitated research in detection of ___pathogens___ in Plants animals environment
26) The DNA fragments are denatured and transferred to _____membrane for analysis. Nitrocellulose
27) An enzyme responsible for removing phosphate groups from many types of molecules___Alkaline Phosphatase
28) **PCR has also facilitated research in detection of ---------- plants, animals and environment**
29) **Type---------- systems are of little value for gene manipulation (Type I**
30) **A suitable system was proposed by ---------- (Smith and Nathans (1973**
31) **Six base recognition site occurs every ---------- bp .(4096)**
32) Colony PCR
33) Long PCR
34) _____is used to amplify one strand of the original DNA more than the other? Asymmetric PCR
35) Is not an advantage of PCR_____ Primer complexities
36) Type I systems were the first to be characterized from *E. coli* _____K12
37) Type II endonucleases recognize and cleave DNA within particular sequences of four to eight nucleotides which have a twofold axis of rotational symmetry i.e. referred as _____*palindromes*
38) _____enzymes have symmetrical recognition sequences Type III
39) Principle of southern blotting?(2)

Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975). Other blotting methods with similar principles, but using protein or RNA, have been named in reference to Edwin Southern's name after. Identify DNA sequence (gene) of interest. Identified DNA may be a small piece of DNA or a mutation

40) What are fragment used in the southern blotting?(2)
BamH1, EcoR1

41) Difference bw type !! And type!!s Type II ? (2)
ANS:Most of the useful R-M system is Type II Type II enzymes recognize a defined sequence and cut within it Type IIs Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness

42) Calculate restriction sites every four, six, eight.?(3)
Four base recognition site occurs every 44 (256) bp Six base recognition site occurs every 46 (4096) bp Eight base recognition site occurs 48 (65,536) bp.

43) What is southern blot and how DNA cleaved? (3)
leukocytes. DNA is cleaved into many fragments by restriction enzyme (e.g, BamH1, EcoR1 etc) . The resulting fragments are separated on the basis of size by electrophoresis. The DNA fragments are denatured and transferred to nitrocellulose membrane for analysis. The labeled probe is added to the blocked membrane in buffer and incubated for several hours to allow the probe molecules to find their targets. Blot is incubated with wash buffers containing NaCl and detergent to wash away excess probe. Radioactive probes enable autoradiographic detection. Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe

44) What is haplotype? (3)
A haploid genotype is a group of alleles inn an organism that are inherited together from a songal parent

45) Write down the procedure for Western blotting?(5)
Procedure
1- Sample preparation
2- Gel Electrophoresis
3- Blotting (or transfer)
4- Blocking
5- Antibody probing
6- Detection

1- The choice of extraction method depends primarily on the sample and whether the analysis is targeting all the proteins in a cell or only a component from a particular subcellular fraction.
2- 2- Samples are loaded into separate wells. A protein marker is also loaded. The separated protein mixtures are transferred to a solid support for further analysis.
3- 3- Transfer can be done in wet or semi-dry conditions. Semi-dry transfer is generally faster. Wet transfer is recommended for large proteins, >100 kD.
4- Blocking is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blotting membrane
5- Protein of interest is detected and localized using a specific antibody. Western blotting protocols utilize a non-labeled primary antibody directed against the target protein. A species-specific, labeled secondary antibody directed against the constant region of the primary antibody is then used. The most common antibody label used in Western blots is HRP.
6- The signal is detected when HRP is exposed to a substrate solution in the final step of the immunodetection procedure.

46) write the Properties of DNA probe ? (5)
PROBES –PROBES: Labeled material to detect a target. For DNA: 20-30 nucleotides, complementary to a region in the gene or DNA. RADIOACTIVE PROBE - P32: Sensitive Relatively cheap Hazardous Radioactive waste disposal regulations should be followed NON-RADIOACTIVE PROBE –BIOTIN: Sensitive Relatively expensive.. HYBRIDIZATION OF PROBES: The binding

between single stranded labeled probe to a complementary nucleotide sequence on the target DNA.. PROBES: Labeled material to detect complementary region in the gene or DNA

47) Write the Nomenclature of restrictions enzyme. ?(5)

Cutting DNA molecules–Nomenclature : A suitable system was proposed by Smith and Nathans (1973). The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet E. coli = Eco, H. influenzae = Hin Lecture No. 36 Strain identification is written as EcoK . In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of H. influenzae, HindI, HindII, HindIII etc. All restriction enzymes have general name endonuclease R and modificationmethylase M followed by the system name, for example, in case of H. influenzae. R. HindIII or M. HindIII Examples of restriction endonuclease nomenclature

48) Difference between all blotting techniques(10)

